

0.1 Simulation Study: Probability-Based National Validation of a Fixed Clinical AI System

0.1.1 Purpose and policy question

This simulation provides a single empirical demonstration of the chapter’s estimation-grounded validation framework. A fixed clinical AI system produces a binary alert for clinical deterioration within 24 hours. Before national deployment, a public authority must determine whether sensitivity is at least $\tau = 0.85$ in the target-country population and whether the same requirement is met among harder-to-predict patients. The complete predictive pipeline, including preprocessing, missing-data handling, fitted parameters, operating threshold, calibration, and post-processing, is locked before the audit is designed.

The experiment evaluates complete estimation strategies

$$\eta_k = (d_k, \hat{\theta}_k, V_k), \quad (1)$$

where d_k is the probability construction process, $\hat{\theta}_k$ the estimator, and V_k the uncertainty procedure. The principal purpose is to show that the evidentiary value of a realised Objective Reference Point (ORP) does not arise from its observations alone. It depends on the target parameter and on whether the estimator and variance procedure are compatible with the construction process.

This is a target-adequacy experiment. The WEIRD-country development setting provides motivation only; source performance and the External Transportability Criterion are not simulated.

0.1.2 Fixed target population and estimands

One finite target population was generated and held fixed throughout the Monte Carlo experiment. It contains all $N_T = 25,585$ eligible admissions in $H = 400$ hospitals during one common, pre-specified audit period. There is no calendar-month sampling stage. Hospitals belong to a frame-known design stratum $H_h \in \{0, 1\}$, with 300 standard hospitals and 100 priority hospitals. The priority stratum has a higher expected prevalence of difficult patients and deterioration events. Patient difficulty is represented by the separate analysis variable $Q_{hi} \in \{0, 1\}$. Both values of Q occur in both hospital strata.

For admission (h, i) , let Y_{hi} indicate true deterioration, A_{hi} denote the alert produced by the locked system, and $T_{hi} = A_{hi}Y_{hi}$ denote a true positive. The national sensitivity estimand is

$$\theta_T = \Pr_T(A = 1 \mid Y = 1) = \frac{\sum_h \sum_i A_{hi} Y_{hi}}{\sum_h \sum_i Y_{hi}}. \quad (2)$$

For $q \in \{0, 1\}$, the patient-subgroup estimands are

$$\theta_{T,q} = \Pr_T(A = 1 \mid Y = 1, Q = q). \quad (3)$$

The finite-population truths were

$$\theta_T = 0.8784, \quad \theta_{T,0} = 0.9480, \quad \theta_{T,1} = 0.7368.$$

National and easier-subgroup performance therefore exceed the adequacy threshold, whereas harder-subgroup performance is clearly below it.

Figure 1 shows why the hospital stratum is useful for prospective enrichment without being identical to the patient subgroup. Priority hospitals contain more harder patients and more outcome events, but sensitivity is also lower in that stratum.

Table 1: Fixed finite-population target quantities.

Estimand	Eligible patients	Positive cases	Event prevalence	Sensitivity
National	25,585	4,819	0.188	0.8784
Easy	20,300	3,231	0.159	0.9480
Hard	5,285	1,588	0.300	0.7368

The harder subgroup is defined at the patient level and occurs in both hospital design strata.

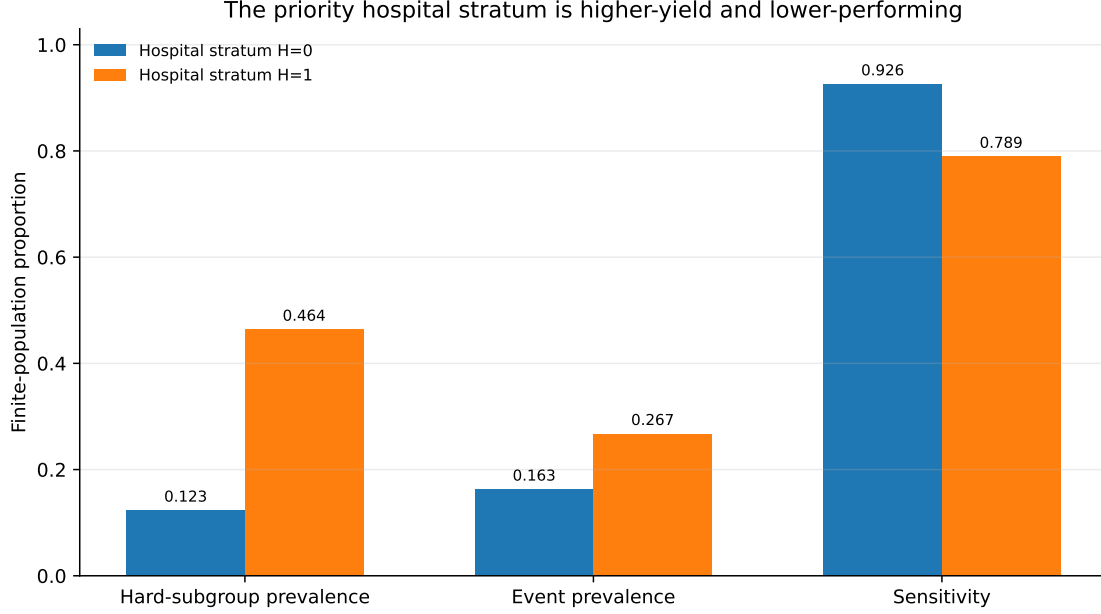


Figure 1: Finite-population structure by hospital design stratum.

0.1.3 Target, observation, and realised ORP distributions

The target distribution P_T is the finite-population empirical distribution assigning equal mass to every eligible admission. For strategy k , the design-induced unweighted observation distribution is

$$P_{\mathcal{O},k}(h,i) = \frac{\pi_{hi}^{(k)}}{\sum_g \sum_r \pi_{gr}^{(k)}}, \quad (4)$$

where $\pi_{hi}^{(k)}$ is the first-order probability that admission (h,i) appears in the ORP. Under hospital sampling followed by a census of eligible admissions, $\pi_{hi}^{(k)} = \pi_h^{(k)}$. The associated unweighted sensitivity parameter is

$$\theta_{\mathcal{O},k} = \frac{\sum_h \sum_i \pi_{hi}^{(k)} A_{hi} Y_{hi}}{\sum_h \sum_i \pi_{hi}^{(k)} Y_{hi}}. \quad (5)$$

This quantity is distinct from both θ_T and the empirical estimate computed from one realised ORP.

In the enriched design, one half of the selected hospitals is drawn from each hospital stratum. Because the frame contains three times as many standard as priority hospitals, the priority-hospital inclusion probability is three times larger. The exact enriched unweighted national parameter is $\theta_{\mathcal{O},3} = 0.8419$, producing the reference-target discrepancy

$$\text{RTD}_3 = \theta_{\mathcal{O},3} - \theta_T = -0.0365. \quad (6)$$

The corresponding discrepancies are -0.0056 for easier patients and -0.0258 for harder patients.

Table 2: Target and design-induced unweighted observation parameters for the enriched ORP.

Estimand	θ_T	$\theta_{O,3}$	RTD
National	0.8784	0.8419	-0.0365
Easy	0.9480	0.9424	-0.0056
Hard	0.7368	0.7110	-0.0258

The inclusion-probability ratio between priority and standard hospitals is three for every enriched sample-size condition, so the observation parameters do not depend on m .

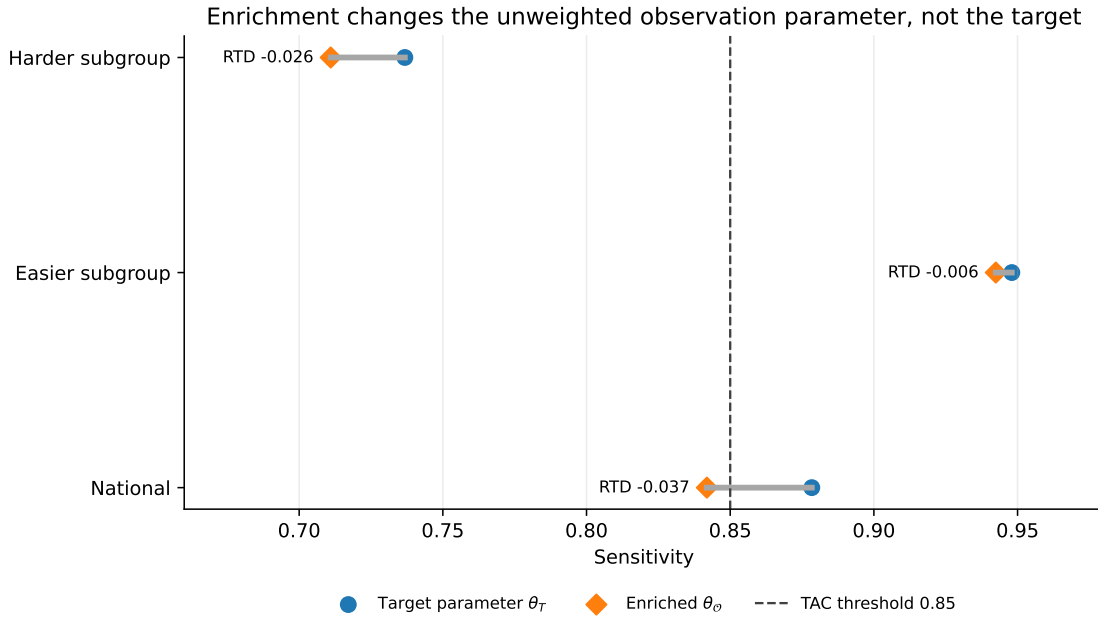


Figure 2: Target parameters and design-induced unweighted observation parameters for the enriched ORP.

0.1.4 Design–estimator–variance strategies

Five strategies were compared. N1 is an ideal patient-level benchmark that requires a complete patient frame. N2 and N4 are operationally realistic hospital-frame designs. N3 and N4 use exactly the same enriched realised ORP within every replication but different point estimators. N4 and N5 use the same ORP and exactly the same point estimate but different uncertainty procedures.

For weighted hospital strategies, stratum totals are estimated independently. For $j \in \{0, 1\}$,

$$\hat{T}_j = \frac{H_j}{m_j} \sum_{h \in s_j} T_h, \quad \hat{Y}_j = \frac{H_j}{m_j} \sum_{h \in s_j} Y_h, \quad (7)$$

and national sensitivity is estimated as

$$\hat{\theta} = \frac{\hat{T}_0 + \hat{T}_1}{\hat{Y}_0 + \hat{Y}_1}. \quad (8)$$

Subgroup estimators use subgroup-specific numerator and denominator totals in exactly the same way. No target positive-case shares are supplied to the estimator.

Table 3: Probability construction, estimator, and uncertainty procedure for the five strategies.

Strategy	Construction process	Point estimator	Variance procedure	Role
N1	Patient SRSWOR	Ordinary sensitivity ratio	Patient-SRS ratio linearisation	Ideal self-weighting benchmark
N2	Proportional stratified hospital SRS	Design-weighted combined ratio	Stratified cluster Taylor linearisation	Realistic target-aligned design
N3	Enriched stratified hospital SRS	Unweighted pooled ratio	Cluster-aware variance around $\theta_{O,3}$	Deliberate estimator mismatch
N4	Same enriched ORP as N3	Design-weighted combined ratio	Stratified cluster Taylor linearisation	Compatible enrichment
N5	Same enriched ORP and point estimate as N4	Design-weighted combined ratio	Naive patient-level variance	Deliberate uncertainty mismatch

The total hospital sample sizes were

$$m \in \{40, 80, 120, 160\}.$$

N2 used proportional allocations (30, 10), (60, 20), (90, 30), and (120, 40). N3–N5 used enriched allocations (20, 20), (40, 40), (60, 60), and (80, 80). Every selected hospital contributed all eligible admissions from the common audit period. N1 patient sample sizes were chosen to equal the expected number of admissions observed under proportional hospital sampling.

N2 and N4 used a stratified hospital-level Taylor linearisation for the ratio. N3 used a cluster-aware variance around the unweighted parameter it actually targeted. N5 instead treated weighted positive patients as independent and unclustered. Logit-transformed 95% confidence intervals were used throughout.

The PR evaluation tolerances were

$$\varepsilon_b = 0.03, \quad \varepsilon_v = \left(\frac{0.03}{1.96} \right)^2. \quad (9)$$

Bias and precision were assessed from 5,000 repeated samples per design and sample-size condition. Empirical interval coverage was reported separately, not folded into the PR variance condition. Because TAC requires a reliable interval in addition to point-estimation PR, a formal TAC classification was issued only when the PR conditions, target-interval validity, and declared variance-procedure compatibility were all satisfied.

0.1.5 Results

Same target parameter under different probability designs. N1, N2, and N4 all recovered national sensitivity when paired with compatible estimators. Across all sample sizes, N1’s absolute national bias was below 0.0004, while N4’s was below 0.0003. N2 was also effectively unbiased. The designs differed in precision because patient SRS and clustered hospital sampling have different variance structures, but all three provided a valid route to the same target parameter once their design information was used.

Table 4: Monte Carlo performance for national sensitivity.

m	Strategy	Bias	ESD	RMSE	Mean SE	Coverage	Evidence
40	N1	+0.0003	0.0140	0.0140	0.0141	0.955	Pass
40	N2	-0.0002	0.0237	0.0237	0.0230	0.925	Fail
40	N3	-0.0365	0.0259	0.0447	0.0249	0.635	Fail
40	N4	+0.0000	0.0212	0.0212	0.0204	0.933	Fail
40	N5	+0.0000	0.0212	0.0212	0.0143	0.817	Fail
80	N1	-0.0000	0.0093	0.0093	0.0094	0.954	Pass
80	N2	+0.0001	0.0157	0.0157	0.0156	0.941	Fail
80	N3	-0.0365	0.0159	0.0398	0.0157	0.290	Fail
80	N4	-0.0001	0.0135	0.0135	0.0135	0.947	Pass
80	N5	-0.0001	0.0135	0.0135	0.0102	0.861	Fail
120	N1	+0.0000	0.0072	0.0072	0.0072	0.949	Pass
120	N2	+0.0000	0.0123	0.0123	0.0119	0.938	Pass
120	N3	-0.0368	0.0111	0.0384	0.0109	0.050	Fail
120	N4	-0.0002	0.0102	0.0102	0.0101	0.942	Pass
120	N5	-0.0002	0.0102	0.0102	0.0083	0.890	Fail
160	N1	+0.0002	0.0057	0.0057	0.0058	0.953	Pass
160	N2	+0.0001	0.0096	0.0096	0.0096	0.946	Pass
160	N3	-0.0365	0.0073	0.0372	0.0073	0.001	Fail
160	N4	+0.0000	0.0078	0.0078	0.0078	0.944	Pass
160	N5	+0.0000	0.0078	0.0078	0.0072	0.928	Fail

Evidence is marked Pass only when the bias and precision tolerances are met, target interval coverage is acceptable, and the declared variance procedure is design-compatible.

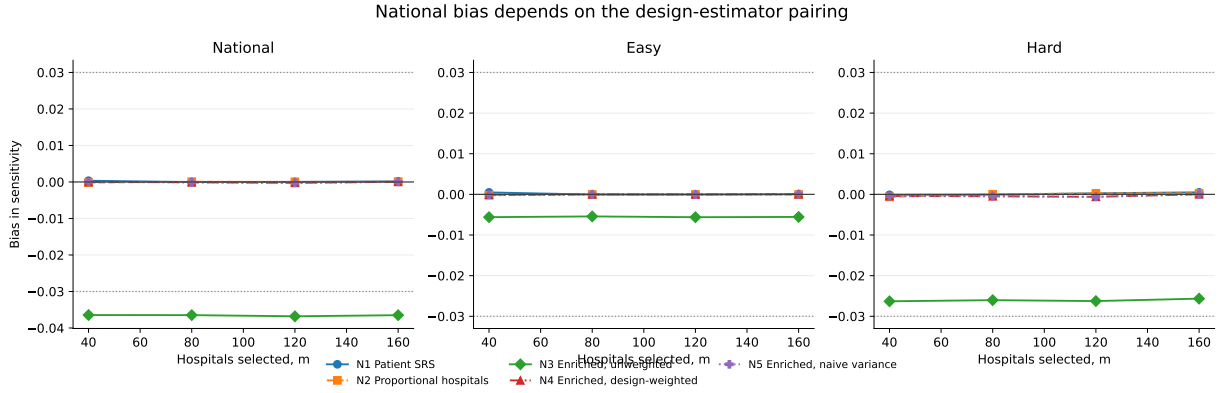


Figure 3: Bias by strategy, hospital sample size, and sensitivity estimand. Dotted horizontal lines mark the pre-specified bias tolerance.

Same enriched ORP, different point estimators. N3 and N4 contained exactly the same selected hospitals and admissions in every replication. Nevertheless, N3’s national bias remained approximately -0.0365 at every sample size, closely matching the exact RTD. Its empirical standard deviation decreased from 0.0259 at $m = 40$ to 0.0073 at $m = 160$, but its mean estimate remained centred near $\theta_{O,3} = 0.8419$. N4 used the same ORP with inverse-inclusion weights and remained centred on $\theta_T = 0.8784$.

This comparison directly separates the realised ORP from the estimation strategy. Weighting did not alter the observed units or transform the ORP into a different dataset. It altered the point estimator’s inferential target.

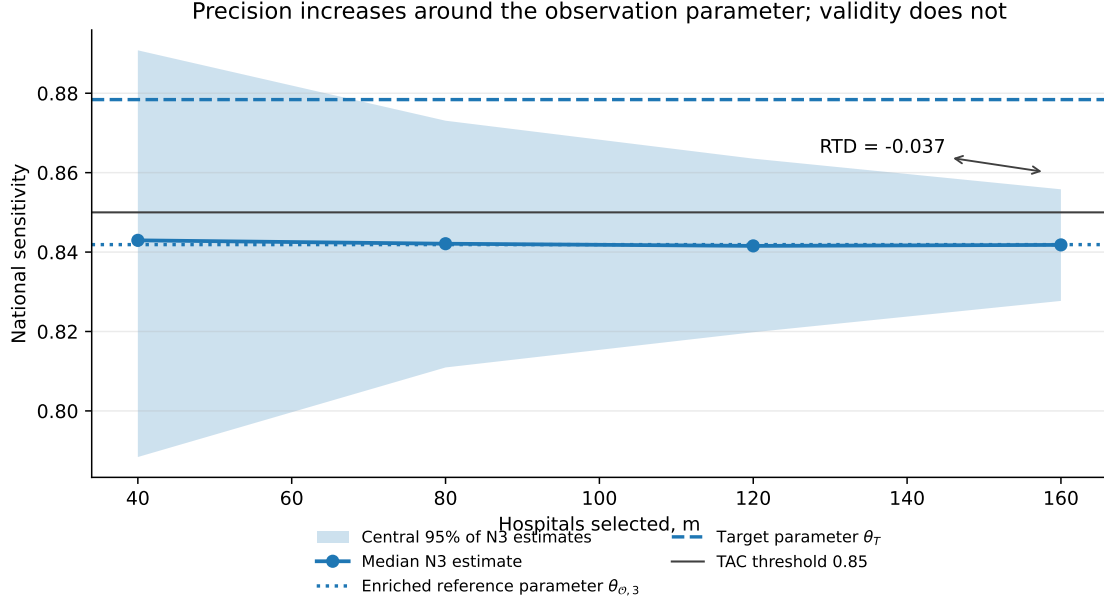


Figure 4: Increasing sample size concentrates N3 around the design-induced observation parameter rather than the national target.

Same ORP and point estimate, different uncertainty. N4 and N5 had numerically identical estimates in every replication. Their uncertainty statements differed substantially. At $m = 80$, the empirical standard deviation of the common national estimator was 0.0135. N4's mean estimated standard error was also 0.0135, with 95% coverage of 0.947. N5's mean standard error was only 0.0102, and coverage fell to 0.861. At $m = 120$, coverage was 0.942 for N4 and 0.890 for N5.

The narrower N5 intervals also produced more mechanically decisive TAC results. For example, at $m = 80$, the mechanical adequate-decision frequency was 68.6% under N5 compared with 49.1% under N4. These additional decisions were not additional evidence; they resulted from ignoring the hospital clustering that generated the ORP.

Table 5: Same enriched samples and point estimates, different uncertainty procedures.

m	Strategy	ESD	Mean SE	SE/ESD	Coverage	Mean width
40	N4	0.0212	0.0204	0.960	0.933	0.0805
40	N5	0.0212	0.0143	0.673	0.817	0.0563
80	N4	0.0135	0.0135	1.000	0.947	0.0530
80	N5	0.0135	0.0102	0.754	0.861	0.0399
120	N4	0.0102	0.0101	0.984	0.942	0.0395
120	N5	0.0102	0.0083	0.811	0.890	0.0326
160	N4	0.0078	0.0078	0.999	0.944	0.0305
160	N5	0.0078	0.0072	0.924	0.928	0.0282

N4 and N5 have numerically identical point estimates in every replication. Differences arise only from the uncertainty procedure.

Enrichment and harder-subgroup evidence. The enriched strategy increased harder-subgroup positive cases by approximately 49% relative to proportional hospital sampling at every sample size.

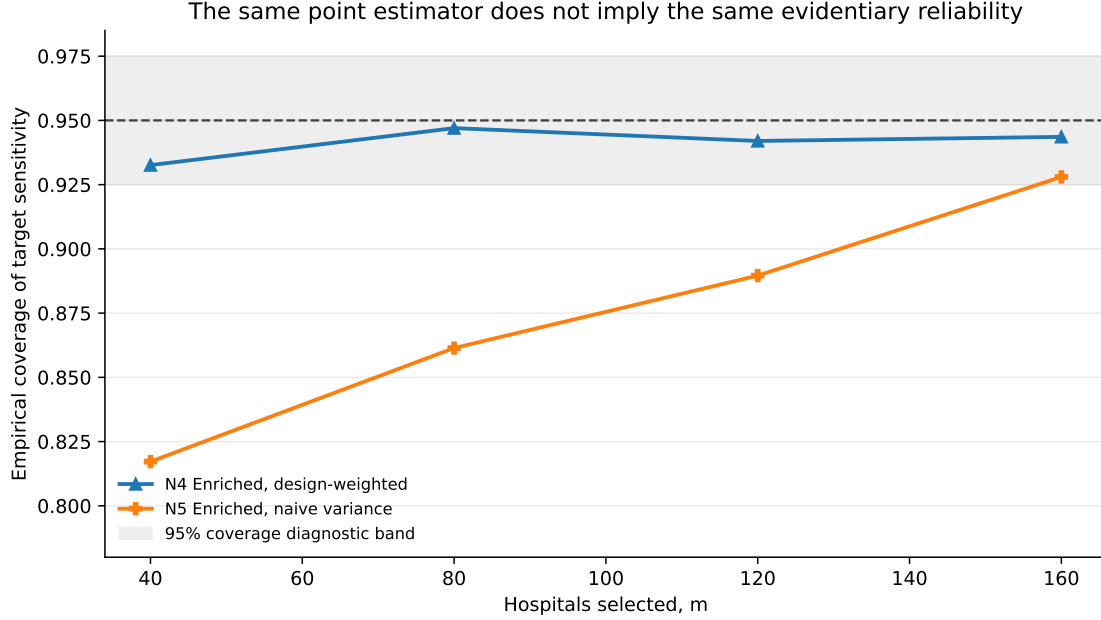


Figure 5: Empirical target-parameter coverage for the same enriched point estimator under compatible and naive variance procedures.

At $m = 160$, the mean count increased from 635 under N2 to 947 under N4. The empirical standard deviation of harder-subgroup sensitivity decreased from 0.0222 to 0.0147, a reduction of about 34%. N4 consequently passed the pre-specified hard-subgroup precision criterion at $m = 160$, while N2 remained evidentially insufficient for that estimand.

Table 6: Effect of enrichment on harder-subgroup evidence.

m	Strategy	Mean hard positives	ESD	RMSE	Precision	Evidence
40	N2	158.9	0.0545	0.0544	Fail	Fail
40	N4	236.3	0.0433	0.0433	Fail	Fail
80	N2	317.1	0.0358	0.0358	Fail	Fail
80	N4	473.2	0.0273	0.0273	Fail	Fail
120	N2	477.1	0.0277	0.0277	Fail	Fail
120	N4	710.5	0.0201	0.0201	Fail	Fail
160	N2	635.2	0.0222	0.0222	Fail	Fail
160	N4	947.0	0.0147	0.0146	Pass	Pass

At $m = 160$, N4 passes the pre-specified precision criterion for harder-subgroup sensitivity, whereas N2 remains evidentially insufficient.

PR was estimand-relative. The same strategy did not receive a single global classification. N4 supported national sensitivity from $m = 80$ onward but supported harder-subgroup sensitivity only at $m = 160$. N2 supported national sensitivity at $m = 120$ and $m = 160$ but did not achieve the required harder-subgroup precision. N3 failed the national bias criterion at every sample size even when its sampling variance became small. These results operationalise the claim that PR is relative to a parameter and tolerance, not a descriptive property of the ORP.

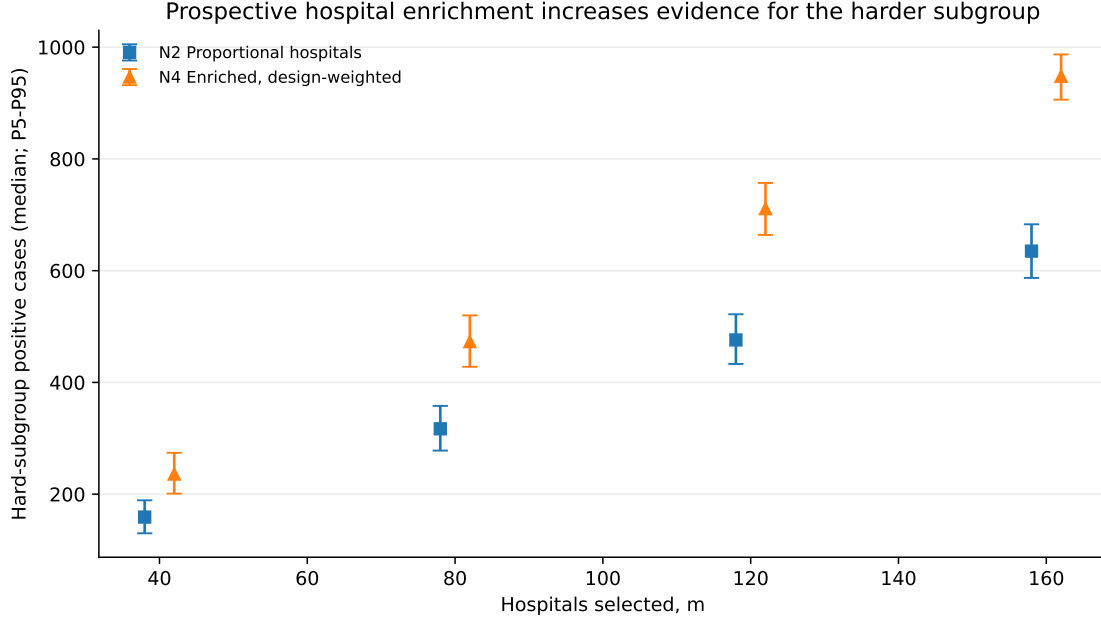


Figure 6: Harder-subgroup positive-case yield under proportional and enriched hospital sampling.

Table 7: Predictive Representativity and uncertainty diagnostics at $m = 160$.

Strategy	Estimand	Bias	Precision	Coverage	V_k compatible	Full evidence
N1	Easy	Pass	Pass	Pass	Pass	Pass
N1	Hard	Pass	Pass	Pass	Pass	Pass
N1	National	Pass	Pass	Pass	Pass	Pass
N2	Easy	Pass	Pass	Pass	Pass	Pass
N2	Hard	Pass	Fail	Pass	Pass	Fail
N2	National	Pass	Pass	Pass	Pass	Pass
N3	Easy	Pass	Pass	Fail	Pass	Fail
N3	Hard	Pass	Pass	Fail	Pass	Fail
N3	National	Fail	Pass	Fail	Pass	Fail
N4	Easy	Pass	Pass	Pass	Pass	Pass
N4	Hard	Pass	Pass	Pass	Pass	Pass
N4	National	Pass	Pass	Pass	Pass	Pass
N5	Easy	Pass	Pass	Pass	Fail	Fail
N5	Hard	Pass	Pass	Pass	Fail	Fail
N5	National	Pass	Pass	Pass	Fail	Fail

TAC followed PR and uncertainty validation. At $m = 160$, N4 produced a formal national TAC decision of Adequate in 92.3% of replications and Inconclusive in 7.7%. For harder-subgroup sensitivity, where the truth was 0.7368, N4 produced Not adequate in 100% of replications. N2 remained evidentially insufficient for the harder-subgroup TAC because its precision criterion failed. N3 was not assigned a national TAC classification because it estimated the wrong national parameter, and N5 was not assigned a formal classification because its declared uncertainty procedure was incompatible with the clustered design.

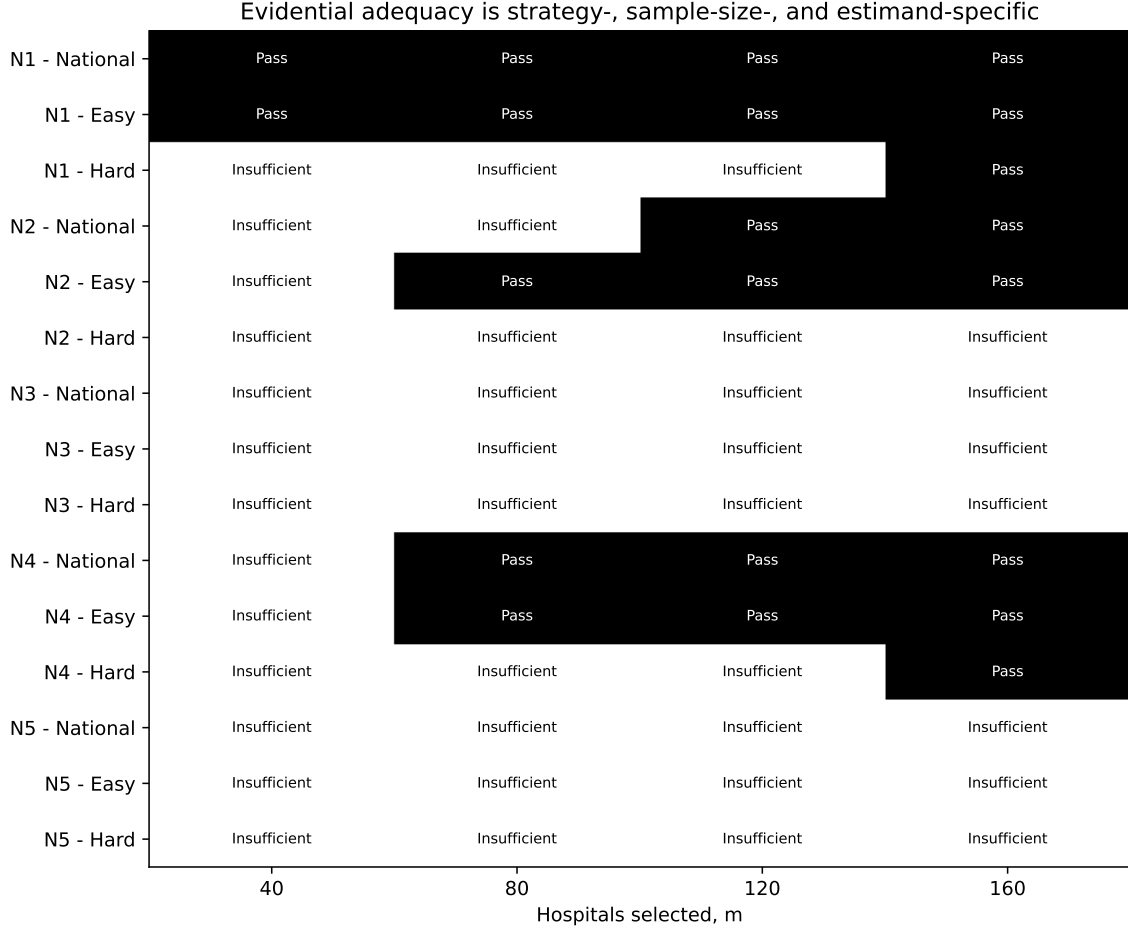


Figure 7: Full evidential status by strategy, sample size, and estimand.

0.1.6 Interpretation

The numerical results support six conclusions. First, P_T defines where the performance claim applies, whereas $P_{O,k}$ describes the unweighted composition induced by the design. Second, the empirical ORP is a realised reference object rather than a target parameter. Third, the same ORP can support different inferential conclusions when paired with different estimators, as demonstrated by N3 and N4. Fourth, the same point estimator can have different evidentiary reliability under different variance procedures, as demonstrated by N4 and N5. Fifth, prospective enrichment can improve harder-subgroup precision while preserving national inference when inclusion probabilities are known and the estimator is compatible. Sixth, increasing sample size reduces sampling error but does not remove a fixed reference–target discrepancy.

The practical implication is that national validation should be planned from the policy question backward. The target population and performance parameter must be specified first; the hospital frame and allocation should then be chosen to support both national and subgroup questions; and the estimator and variance procedure must be declared as part of the strategy. PR is the evidential gatekeeper. TAC is the subsequent decision rule and cannot repair a failure of target alignment or uncertainty validity.

The conclusions concern sensitivity at one locked operating threshold. They do not establish

Table 8: Formal TAC decision frequencies at $m = 160$.

Strategy	Estimand	Adequate	Inconclusive	Not adequate	Evidentially insufficient
N1	Easy	1.000	0.000	0.000	0.000
N1	Hard	0.000	0.000	1.000	0.000
N1	National	0.998	0.002	0.000	0.000
N2	Easy	1.000	0.000	0.000	0.000
N2	Hard	0.000	0.000	0.000	1.000
N2	National	0.783	0.217	0.000	0.000
N3	Easy	0.000	0.000	0.000	1.000
N3	Hard	0.000	0.000	0.000	1.000
N3	National	0.000	0.000	0.000	1.000
N4	Easy	1.000	0.000	0.000	0.000
N4	Hard	0.000	0.000	1.000	0.000
N4	National	0.923	0.077	0.000	0.000
N5	Easy	0.000	0.000	0.000	1.000
N5	Hard	0.000	0.000	0.000	1.000
N5	National	0.000	0.000	0.000	1.000

Mechanical threshold results are suppressed whenever the full evidential gate fails.

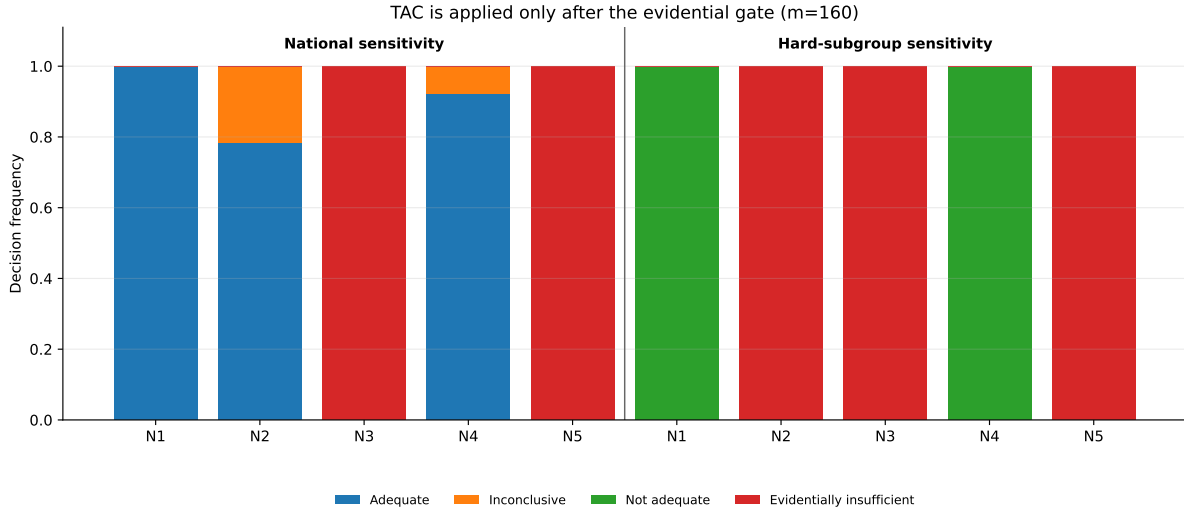


Figure 8: Formal TAC decision frequencies at $m = 160$, with evidential insufficiency shown separately from interval-based inconclusiveness.

global model safety, calibration, discrimination, or clinical utility, each of which would require its own estimand-compatible validation strategy.

.1 Technical Appendix to the National Probability-Based Validation Simulation

.1.1 Finite-population generation

The finite population contains $H = 400$ hospitals, of which $H_0 = 300$ belong to the standard design stratum and $H_1 = 100$ to the priority stratum. Hospital sizes are generated from a truncated

log-normal construction,

$$N_h = \min\{105, \max\{35, \text{round}(\tilde{N}_h)\}\}, \quad \log \tilde{N}_h \sim \mathcal{N}(4.1276331, 0.25^2).$$

The realised total is $N_T = 25,585$.

For each hospital, let

$$(b_{Qh}, b_{Yh}, b_{Ah})^\top \sim \mathcal{N}(0, \Sigma_b),$$

with standard deviations (0.30, 0.38, 1.20) and correlation matrix

$$R_b = \begin{pmatrix} 1 & 0.15 & 0 \\ 0.15 & 1 & 0.15 \\ 0 & 0.15 & 1 \end{pmatrix}.$$

For patient i in hospital h , the harder-subgroup indicator is

$$\text{logit Pr}(Q_{hi} = 1 \mid H_h, b_{Qh}) = \text{logit}(0.12) + 1.80H_h + b_{Qh}. \quad (10)$$

The deterioration outcome is generated from

$$\text{logit Pr}(Y_{hi} = 1 \mid H_h, Q_{hi}, b_{Yh}) = \text{logit}(0.14) + 0.45H_h + 0.65Q_{hi} + b_{Yh}. \quad (11)$$

Among outcome-positive admissions, the locked alert is generated from

$$\text{logit Pr}(A_{hi} = 1 \mid Y_{hi} = 1, H_h, Q_{hi}, b_{Ah}) = \text{logit}(0.97) - 0.55H_h - 2.00Q_{hi} + b_{Ah}. \quad (12)$$

For completeness, false-alert probabilities among $Y = 0$ are generated from

$$\text{logit Pr}(A_{hi} = 1 \mid Y_{hi} = 0, H_h, Q_{hi}, b_{Yh}) = \text{logit}(0.10) + 0.15H_h + 0.15Q_{hi} + 0.20b_{Yh}.$$

False alerts do not enter the sensitivity estimands.

The population is generated once with master seed 141421 and then frozen. No DGP parameter is altered after observing the strategy results.

.1.2 Target and observation parameters

The finite-population target distribution assigns mass $1/N_T$ to every eligible admission. The national and subgroup parameters are ratios of fixed finite-population totals:

$$\theta_{T,q} = \frac{T_{T,q}}{Y_{T,q}}, \quad T_{T,q} = \sum_h \sum_i \mathbf{1}(Q_{hi} = q) A_{hi} Y_{hi}, \quad Y_{T,q} = \sum_h \sum_i \mathbf{1}(Q_{hi} = q) Y_{hi}.$$

The national parameter is obtained by omitting the subgroup indicator.

For design k , the unweighted observation distribution is

$$P_{\mathcal{O},k}(h, i) = \frac{\pi_{hi}^{(k)}}{\sum_g \sum_r \pi_{gr}^{(k)}}.$$

Its subgroup sensitivity parameter is

$$\theta_{\mathcal{O},k,q} = \frac{\sum_h \sum_i \pi_{hi}^{(k)} \mathbf{1}(Q_{hi} = q) A_{hi} Y_{hi}}{\sum_h \sum_i \pi_{hi}^{(k)} \mathbf{1}(Q_{hi} = q) Y_{hi}}.$$

These parameters were calculated exactly from the locked finite population before Monte Carlo sampling.

.1.3 Inclusion probabilities and allocations

N1 draws a patient SRSWOR of size

$$n_m = \text{round} \left(N_T \frac{m}{400} \right),$$

which gives $n_m \in \{2559, 5117, 7676, 10234\}$ for $m \in \{40, 80, 120, 160\}$. Every patient has inclusion probability n_m/N_T .

For hospital strategies, hospitals are sampled independently by SRSWOR within stratum. If m_j hospitals are selected from H_j , then every admission in hospital stratum j has first-order inclusion probability

$$\pi_{hi} = \pi_h = \frac{m_j}{H_j}.$$

N2 uses proportional allocations

$$(m_0, m_1) \in \{(30, 10), (60, 20), (90, 30), (120, 40)\}.$$

N3–N5 use enriched allocations

$$(m_0, m_1) \in \{(20, 20), (40, 40), (60, 60), (80, 80)\}.$$

All admissions in selected hospitals are observed during the same fixed audit period. There is no patient subsampling inside hospitals and no month-level selection stage.

.1.4 Estimators

For N1, ordinary sample sensitivity is

$$\hat{\theta}_{N1} = \frac{\sum_{i \in s} A_i Y_i}{\sum_{i \in s} Y_i}.$$

For N3, the enriched ORP is analysed using the pooled unweighted ratio

$$\hat{\theta}_{N3} = \frac{\sum_{h \in s_0 \cup s_1} T_h}{\sum_{h \in s_0 \cup s_1} Y_h}.$$

For N2, N4, and N5, define design-expanded stratum totals

$$\hat{T}_j = \frac{H_j}{m_j} \sum_{h \in s_j} T_h, \quad \hat{Y}_j = \frac{H_j}{m_j} \sum_{h \in s_j} Y_h.$$

The design-weighted combined ratio is

$$\hat{\theta}_w = \frac{\hat{T}_0 + \hat{T}_1}{\hat{Y}_0 + \hat{Y}_1}.$$

For subgroup q , T_h and Y_h are replaced by T_{hq} and Y_{hq} before stratum totals are added.

.1.5 Variance procedures

For the design-weighted combined ratio, let

$$z_{hj} = T_{hj} - \hat{\theta}_w Y_{hj}$$

and let $s_{z,j}^2$ be the sample variance of z_{hj} among selected hospitals in stratum j . The Taylor variance estimator is

$$\widehat{\text{Var}}(\hat{\theta}_w) = \frac{\sum_{j=0}^1 H_j^2 \left(1 - \frac{m_j}{H_j}\right) \frac{s_{z,j}^2}{m_j}}{(\hat{Y}_0 + \hat{Y}_1)^2}. \quad (13)$$

Independent stratum contributions are therefore added before division by the squared expanded denominator.

For N3, the same ratio-linearisation idea is applied to unexpanded sample sums. Its variance estimates sampling dispersion around the parameter targeted by the unweighted estimator:

$$\widehat{\text{Var}}(\hat{\theta}_{N3}) = \frac{\sum_{j=0}^1 m_j \left(1 - \frac{m_j}{H_j}\right) s_{z,j}^2}{\left(\sum_{h \in s_0 \cup s_1} Y_h\right)^2}.$$

N5 uses the same weighted point estimate as N4 but treats positive patients as independent. With stratum weights $w_j = H_j/m_j$, its naive variance is

$$\widehat{\text{Var}}_{\text{naive}}(\hat{\theta}_w) = \frac{\sum_{j=0}^1 w_j^2 \left[T_{s,j} (1 - \hat{\theta}_w)^2 + (Y_{s,j} - T_{s,j}) \hat{\theta}_w^2 \right]}{\left(\sum_{j=0}^1 w_j Y_{s,j} \right)^2}.$$

This expression ignores the hospital-level dependence and is intentionally incompatible with the design.

For all strategies, a logit-transformed 95% interval is formed as

$$\text{expit} \left[\text{logit}(\hat{\theta}) \pm 1.96 \frac{\widehat{\text{SE}}(\hat{\theta})}{\hat{\theta}(1 - \hat{\theta})} \right].$$

.1.6 PR, uncertainty, and TAC classification

For each strategy and estimand, the Monte Carlo bias and variance are

$$\text{Bias}_k = \frac{1}{R} \sum_{r=1}^R \hat{\theta}_{kr} - \theta_T, \quad \text{Var}_k = \frac{1}{R-1} \sum_{r=1}^R (\hat{\theta}_{kr} - \bar{\theta}_k)^2.$$

The declared point-estimation PR conditions are

$$|\text{Bias}_k| \leq 0.03, \quad \text{Var}_k \leq (0.03/1.96)^2.$$

Interval coverage is reported separately. Formal TAC requires the PR point-estimation conditions, acceptable target-parameter coverage, and a variance procedure structurally compatible with the design. The last condition prevents accidental coverage in one simulation cell from legitimising N5's known incompatible variance rule.

For a valid interval $[L, U]$,

$$\text{TAC}(0.85) = \begin{cases} \text{Adequate}, & L \geq 0.85, \\ \text{Not adequate}, & U < 0.85, \\ \text{Inconclusive}, & L < 0.85 \leq U. \end{cases}$$

When the evidential gate fails, the formal status is Evidentially insufficient. Mechanical interval-threshold outcomes are retained in the replication file for diagnostic purposes only.

.1.7 Monte Carlo uncertainty

For R estimable replications, the Monte Carlo standard error of bias is

$$\text{MCSE}(\widehat{\text{Bias}}) = \frac{\text{ESD}(\widehat{\theta})}{\sqrt{R}}.$$

For an estimated proportion $\widehat{p}$, including coverage, TAC, and non-estimability frequencies,

$$\text{MCSE}(\widehat{p}) = \sqrt{\frac{\widehat{p}(1 - \widehat{p})}{R}}.$$

Writing $e_r^2 = (\widehat{\theta}_r - \theta_T)^2$, the delta-method MCSE of RMSE is

$$\text{MCSE}(\widehat{\text{RMSE}}) = \frac{\text{SD}(e_r^2)}{2\widehat{\text{RMSE}}\sqrt{R}}.$$

.1.8 Seed mapping and implementation checks

The Python sub-seed for master seed M , design label L , sample size m , replication r , and language tag is generated from

$$\text{SHA256}(M \mid L \mid m \mid r \mid \text{python}).$$

The first eight digest bytes are interpreted as an unsigned little-endian integer and reduced modulo $2^{63} - 1$. N3–N5 share the label **ENRICHED**, ensuring identical samples within replication. The R implementation uses the analogous SHA-256 mapping with language tag **R** and an R-compatible integer reduction.

The executable validation suite verifies inclusion probabilities, absence of month sampling, exact sample identity for N3–N5, exact point-estimate identity for N4 and N5, direct enumeration of finite-population truths, approximate unbiasedness of N1, design consistency of N4, convergence of N3 toward $\theta_{\mathcal{O},3}$, detection of N5 uncertainty failure, and the absence of formal TAC classifications after an evidential-gate failure. The full machine-readable results and software checksums are stored in `results/summary/validation.results.json`.